

V41 CALIBRATION — FAILURE ANALYSIS

Drift–Sense Phase 2 Submission

Purpose. V41 is the conservative calibration build. The established localization, pose, and binary *found* decisions are preserved; the principal change is confidence scoring.

Confidence rule. $S = 0.90 S_{\text{raw}} + 0.05 S_{\text{top1}} + 0.05 S_{\text{corr}}$. For *found*=0, the raw term is zero while the remaining evidence is retained, avoiding a flat zero-confidence bin.

1. FAILURE MODES

Failure mode	Mechanism
Periodic replica confusion	Repeated semiconductor structures produce strong correlation peaks at incorrect lattice replicas; a strong local maximum can therefore be spatially wrong.
Degraded true-instance miss	Charging, scan distortion, defocus and elevated noise can suppress the true response, causing a present reference to be rejected.
Absent-image false positive	The absent set contains plausible periodic distractors but no true instance; an unconditional maximum can become a confident wrong grab.
Rejected-score collapse	Assigning every <i>found</i> =0 pair score zero destroys ordering information between correct rejections and degraded false negatives.
Incorrect-site pose	Scale/rotation attached to a wrong replica has no useful physical meaning; pose must be interpreted with localization correctness.

2. V41 MITIGATION

- **Protect localization:** retain the established localization path; do not introduce an unvalidated replacement ranker.
- **Protect pose:** retain the validated scale/rotation estimation.
- **Protect rejection:** preserve the established binary *found* decision.
- **Restore confidence ordering:** retain weak evidence from rejected pairs through the V41-F residual mix.
- **Keep inference simple:** deterministic and interpretable; no runtime training or network access.

3. CALIBRATION INSIGHT

The key observation concerns the $found=0$ population. Correct absent-image rejections can still contain moderate distractor evidence, while degraded present pairs can have extremely weak evidence. Assigning a flat zero to every rejected pair therefore destroys useful confidence ordering.

V41 retains this information with $S_{\text{reject}} = 0.05 S_{\text{top1}} + 0.05 S_{\text{corr}}$. The modification is deliberately simple so that confidence calibration does not independently move the established localization decision.

4. DEVELOPMENT EVIDENCE

Metric	Baseline	V41	Change
Spearman rank correlation	0.5995	0.6159	+0.0164
ROC AUC (diagnostic)	0.8248	0.8690	+0.0442
Brier score (diagnostic)	—	0.2226	—

Metric status. These are development/diagnostic values from the V41 artifact, not the hidden organizer score. The competition specification defines the calibration component using AUC; Spearman is supplementary evidence.

5. REMAINING RISKS AND CONTROLS

Risk	Control
Wrong periodic replica	Keep the established localization decision protected; calibration does not independently move the coordinate.
False negative on degraded data	Do not lower the binary rejection threshold solely to improve calibration.
Calibration overfitting	Treat development metrics as evidence, not a guarantee of hidden-test performance.
Runtime failure	Package every runtime dependency/model locally and test from a clean directory.
Schema failure	Verify one row per input pair and enforce zero pose values whenever $found=0$.

6. SUBMISSION INTEGRITY CHECKLIST

1. `python register.py --input pairs.csv --output predictions.csv` executes successfully.
2. Every input `pair_id` appears exactly once.
3. Output columns are exactly `pair_id`, `x`, `y`, `theta`, `scale`, `found`, `score`.
4. For every $found=0$: $x=0$, $y=0$, $theta=0$, $scale=0$.
5. All runtime modules, model files and configurations are inside the ZIP.
6. No network access, external paths or development caches are required.
7. The ZIP runs from a clean directory without the Git repository.

FINAL POLICY

V41 is the safe fallback submission: improve confidence ordering while protecting established localization, pose, and rejection behavior. The dominant remaining risk is periodic replica ambiguity; V41's calibration change does not claim to solve that localization problem.